

108 AI — Desktop Client Master Plan

****Documento unificato**** · **Ultima revisione:** ****2026-06-16**** **Consolida e allinea:**
`desktop-agent-roadmap-complete.md`, `phase-5-desktop-agent-v2.md`, `desktop-agent-installer-plan.md`,
`tracks/108-ai/108AI-Desktop-Bridge.md` ****Stato codice verificato**** su `aia-platform/apps/local-agent`,
`packages/desktop-bridge`, `apps/client`, `apps/dashboard`

1. Executive Summary

Il Desktop Client 108 AI non è un singolo binario: è un sistema a 4 layer che lavorano insieme.

Layer	Package / App	Ruolo
L1 — Agent runtime	apps/local-agent (@108ai/desktop v0.3.0)	CLI, shell REPL, WebSocket verso gateway, triage/jobs, extensions
L2 — Native bridge	packages/desktop-bridge	Percezione/azione OS (Windows primario, macOS parziale)
L3 — Control plane web	apps/client + apps/dashboard	Download, monitor connessione, approvazioni consulente
L4 — Cloud	apps/gateway + LiteLLM	Decisioni AI, billing multi-tenant, policy

Principio architetturale (invariato):

*L'intelligenza vive nel cloud. Il desktop esegue localmente. Il gateway decide ****cosa**** fare; il client decide ****se**** farlo (veto + permission system).*

Maturità complessiva stimata: ~75–80% del piano funzionale agent/CLI · ~40% del piano prodotto installabile · ~85% del piano coding assistant (Phase 5 / Fase B chiusa) · ~70% del bridge nativo Windows.

2. Architettura (vista unificata)

```
flowchart TB
  subgraph Cloud["Cloud (Gateway + LiteLLM)"]
    GW[Gateway API]
    LLM[LiteLLM]
  end

  subgraph Desktop["Desktop Client"]
    AGENT["local-agent\n108ai CLI + shell"]
    BRIDGE["desktop-bridge\nUIA / screen / input"]
    TRAY["Tray + updater stub"]
    EXT["extensions/\nMCP · skills · agents"]
    UI_LOCAL["Web UI locale\n127.0.0.1:7891"]
  end

  subgraph Web["Web Apps"]
    CLIENT["client/desktop-agent"]
    DASH["dashboard/desktop/*"]
  end

  GW <-->|WebSocket| AGENT
  AGENT --> BRIDGE
  AGENT --> EXT
  AGENT --> UI_LOCAL
  CLIENT -->|download / status| GW
  DASH -->|monitor / approve| GW
  AGENT -->|OAuth browser| GW
```

3. Matrice stato reale (codice vs piano)

Legenda: implementato e wired · parziale / stub · non iniziato · solo in documentazione

3.1 Agent runtime — Sprint roadmap (`local-agent`)

Area	Piano (roadmap)	Stato codice	Evidenza
Sprint 1 Fondamenta		local-router, script-store, local-cache, shell.ts REPL	
Sprint 2 Browser/Email/Calendar		integrations/chrome, gmail, calendar, imap-client	
Sprint 3 Office/OCR/Keys		office-*, vision-llm, provider-keys	
Sprint 4 Desktop + Multi-agent		capabilities/desktop.ts, multi-agent/	
Sprint 4b Messaging		telegram-bot, whatsapp-*	
Sprint 5 Business Italia		Solo entry: fatture-in-cloud.ts + triage billing	
Sprint 6 Hardening		hardening/*, SSE, token budget, rate limit tenant	
Sprint 7 Documentazione		USER-GUIDE, SECURITY-RUNBOOK, ADR-001; manca set ADR completo	
Sprint 8 Triage		triage/engine, scheduler, CLI in shell	
Sprint 9 Jobs		jobs/* (store, executor, scheduler, templates)	
Sprint 10 Resources		resources/monitor, auto-healer	
Sprint 11 Extensibility		extensions/* completo; gap residui sotto	
TS strict tsc --noEmit	—	Verificato 2026-06-16	
Vitest	—	65 test	

Gap Sprint 11 (residui verificati):

Item	Stato	Note
MCP stdio + SSE/HTTP		extensions/mcp/
MCP tools in agent loop		extensions/agents/mcp-tools.ts
mcp install		npm/git/preset in extensions/mcp/install.ts

MCP usage metrics | | extensions/mcp/usage.ts
Store catalog locale | | extensions/ui/store/catalog.ts
Store catalog online fetch | | storeCatalogOnlineUrl + cache JSON
Store install da online | | store/installer.ts + API web
Install wizard | | config.ts runSetupWizard() — campi base
Agent switcher | | CLI + web API /api/agent/use; no Electron/Tauri
Sandbox shell in jobs | | jobs/executor.ts + shell-security
/triage, /job come YAML commands | | builtin: in ~/.108ai/commands/

3.2 Coding capabilities — Phase 5 (`phase-5-desktop-agent-v2.md`)

Capability	Piano	Stato codice	Note
shell.execute	v0.2		capabilities/shell.ts + registry
shell.executeStream	v0.2		capabilities/shell.ts + shell-process.ts + WS shell.stream
shell.terminate / getRunning	v0.2		capabilities/shell.ts
code.edit / code.editMulti	v0.2		capabilities/code.ts (alias su filesystem)
code.write / readRange	v0.2		capabilities/code.ts
git.*	v0.2		capabilities/git.ts (push/reset gated)
search.grep / glob / find	v0.3		capabilities/search.ts
web.fetch / web.search	v0.3		capabilities/web.ts + SSRF guard
process.*	v0.3		capabilities/process.ts
mcp.* (capability WS)	v0.4		Implementato come extension /mcp, non action mcp.callTool
PTY / terminal streaming	v0.4		
index.* local RAG	v0.5		embeddings-cache.ts + extensions/knowledge/ (lite, no sqlite-vec)
context.assemble	v0.5		
Rebranding @108ai/desktop	v1.0		package.json v0.3.0
Tray icon / splash	v1.0		tray.ts usa fallback; no icona nativa Win
Installer MSI/DMG/deb			Beta: pnpm package:beta + install-108ai.ps1 (Fase E: MSI/Tauri)
Tauri/Electron shell	v1.0 opt		Decisione: Tauri v2; Fase E

Azioni registrate oggi (capabilities/index.ts): include shell., git., code., search., web., process. + filesystem/desktop.
Target ≥35: raggiunto [verificato].

3.3 Installer & prodotto installabile (`desktop-agent-installer-plan.md`)

Feature	Piano v1.0	Stato codice	File
Binary compilato (esbuild)			scripts/build.mjs
108ai --install PATH			cli.ts handleInstall()
CLI one-shot / pipe			cli.ts
Shell REPL			shell.ts
Background agent + WS			index.ts, connection.ts
OAuth browser login			auth.ts
Installer idempotente (Repair/Update/Uninstall UI)	Target		--install copia; no menu primo avvio
autostart.ts OS startup	Target		Non esiste
System tray menu completo	Target		tray.ts — stati OK; menu limitato, node-notifier non tray nativa
Auto-update funzionale	Target		updater.ts presente; CURRENT_VERSION hardcoded 0.2.0 (drift vs package 0.3.0)
Download da gateway	Target		gateway/routes/desktop-agent-download.ts
Raccomandazione Electron-less v1	ADR implicita		Nessun Electron in repo

3.4 Desktop Bridge (`108AI-Desktop-Bridge.md`)

Area	Piano	Stato
12 azioni desktop (click, type, screenshot...)	Fase 3.5	Windows via @aia/desktop-bridge

Window guard + confirmation | Sicurezza | desktop-bridge/safety/
Dashboard consulente (monitor, screenshot) | Fase 3.5 | dashboard/.../desktop/*
Client download page | — | client/routes/desktop-agent.tsx
macOS client nativo | Fase 8 | Pianificato
Multi-monitor avanzato | Roadmap | Limitazioni documentate

3.5 UI React — Triage & Jobs (gap cross-doc)

Componente	Dove nel piano	Stato
Triage panel React	roadmap Sprint 8/9	TriagePanel.tsx
Job list/detail React	roadmap Sprint 9	JobListPanel.tsx
Web UI locale agent	Sprint 11	extensions/ui/server.ts :7891
Dashboard desktop monitor	Bridge doc	DesktopMonitor, DesktopActionCard

4. Registro drift documentale

Documenti che non riflettono lo stato codice al 2026-06-16 — da aggiornare o trattare questo file come fonte di verità.

Documento	Drift	Azione
desktop-agent-roadmap-complete.md	Header Sprint 6/11 ma sezione gap Sprint 11 ancora lista item già fatti (MCP metrics, online catalog)	Alli
phase-5-desktop-agent-v2.md	Stato reale aggiornato: v0.2/v0.3 implementati, MCP come extension	Allineato (2026-06-17)
desktop-agent-installer-plan.md	Dice "updater stub non funzionale" — codice ha updater.ts quasi completo ma non wired + version drift	Aggiorna
local-agent/README.md	Elenca code.*, git.*, web.* come se esistessero	Allineare a capabilities/index.ts
updater.ts	CURRENT_VERSION = '0.2.0' vs package.json 0.3.0	Fix tecnico minore

5. Roadmap unificata per fasi

Fase A — Beta PMI · ****CHIUSA**** (2026-06-17)

Obiettivo: *"installa, login, triage mattutino, job automatici, consulente vede tutto"* — ***go/no-go beta soddisfatto***.

#	Deliverable	Track	Stato
A1	Installer idempotente + autostart Win/Mac/Linux	Installer	installer.ts, autostart.ts, paths.ts
A2	Tray nativa (systray2) + menu Shell/Dashboard/Pause/Settings/Quit/Update	Installer	tray.ts + fallback node-notifier
A3	Wire auto-update + sync versione package	Installer	version.ts, updater.ts + menu "Riavvia per aggiornare"
A4	First-run: triage schedulato lun-ven 07:00 + notify desktop	Installer	first-run.ts
A5	Schedulers triage/job in 108ai agent (non solo shell)	Runtime	index.ts
A6	UI React: Triage summary panel (dashboard)	UI	TriagePanel.tsx + gateway /triage
A7	UI React: Job list panel (dashboard)	UI	JobListPanel.tsx + gateway /jobs
A8	Pacchetto beta Windows (exe + install-108ai.ps1)	Packaging	scripts/package-beta.mjs

Go/no-go beta: tsc verde · test Vitest · installer Win10+ · OAuth OK · triage schedulato in agent mode · audit JSONL.

Fuori scope Fase A (Fase E): MSI firmato / DMG notarized — distribuzione via zip + PowerShell installer per beta PMI.

Fase B — Coding assistant competitivo (6–8 settimane) ****CHIUSA**** (2026-06-16)

Obiettivo: parità funzionale con Claude Code su shell/git/search.

| Deliverable | Stato

- B1 | git.* (status/diff/log/commit/branch/stash/blame + policy push/reset) | capabilities/git.ts
- B2 | shell.executeStream + process registry + terminate/getRunning | shell-process.ts
- B3 | Namespace code.* (readRange/edit/editMulti/write) | capabilities/code.ts
- B4 | web.fetch con blacklist IP privati + web.search (Brave) | capabilities/web.ts
- B5 | search.* unificato (grep/glob/find) | capabilities/search.ts
- B6 | Slash /model + /clear in shell REPL | shell.ts

Config: shellEnabled, gitEnabled, gitAllowPush, gitAllowDestructive in ~/.108ai/config.json.

Security: tutte le azioni registrate in security.ts + SSRF su web.fetch.

Streaming: eventi shell.stream via WebSocket quando agent mode attivo.

Fase C — Extensibility & store (3–4 settimane) **CHIUSA** (2026-06-16)

| Deliverable | Stato

- C1 | mcp install generico npm/git | extensions/mcp/install.ts
- C2 | Store online → install flow end-to-end | store/installer.ts + /api/store/install
- C3 | Migrazione /trriage, /job → YAML commands | (già in Fase A/B)
- C4 | Sandbox shell enforcement in jobs/executor.ts | validateShellCommand + permissions
- C5 | Firma autore pacchetti store | security/store-signature.ts (HMAC)

Store install: CLI /ui store install <id> · Web UI tab Store · firma HMAC con AIA_STORE_SIGNING_KEY.

MCP install: /mcp install npm <pkg> · /mcp install git <url> --command node --args dist/index.js.

Fase D — Business Italia completo (Sprint 5)

| Deliverable

- D1 | PSD2 / open banking adapter
- D2 | Tracking corrieri (BRT/GLS/SDA)
- D3 | WooCommerce / Shopify
- D4 | Firma digitale + PEC automatica

Fase E — GUI nativa v2 (opzionale, post-feedback)

Decisione documentata: Tauri (non Electron) per chat locale, permission dialog, terminal integrato.

Riferimento: phase-5-desktop-agent-v2.md §5.15, desktop-agent-installer-plan.md Opzione B v2.0.

Fase F — macOS & bridge (Fase 8 Bridge doc)

Parità Windows su percezione/azione; LaunchAgent autostart; notarization.

6. Metriche di successo (consolidate)

Metrica | Target v1.0 | Stato attuale

- Azioni WS registrate | ≥ 35 | ~28 [verificato]
- tsc --noEmit | 0 errori |
- Test Vitest | ≥ 30 | 34

Crash rate sessioni | < 0.1% | [ignoto] — no telemetria prod
Install size binary | < 50 MB | [non verificato] — misurare post-build
Memory idle | < 100 MB | [non verificato]
Auto-reconnect 60s | > 99% | Implementato in connection.ts [probabile]
Audit coverage azioni | 100% | security.ts auditLog

7. Mappa documenti sorgente

Documento | Scope | Usare per

Questo file | Stato unificato + priorità | Decisioni prodotto, sprint planning
desktop-agent-roadmap-complete.md | Sprint 1–11 dettaglio ore/costi | Storia implementazione, task granulari
phase-5-desktop-agent-v2.md | Coding assistant API design | Spec shell.* git.* mcp.*
desktop-agent-installer-plan.md | Prodotto installabile | Tray, autostart, updater, shell UX
tracks/108-ai/108AI-Desktop-Bridge.md | Visione business + bridge | Vendita PMI, casi d'uso, macOS
coding-agent-capabilities.md | Capabilities reference | Gateway ↔ agent contract
apps/local-agent/docs/USER-GUIDE.md | Operatività utente | Beta esterni
apps/local-agent/docs/SECURITY-RUNBOOK.md | Incident response | Compliance

8. Prossimi 3 passi consigliati (decisione Elios)

- Fase A1–A3 — Rendere il prodotto installabile senza competenze tecniche (ROI immediato per PMI).
- Fix drift — updater.ts version + README capabilities + snellire gap table in roadmap.
- Fase A6–A7 — Due pannelli React in dashboard (triage + jobs): sbloccano il consulente senza Tauri.

9. Changelog documento

Data | Revisione

2026-06-16 | Creazione master plan; verifica codice local-agent v0.3.0, bridge, client, dashboard

108 Vision — Costruiamo la direzione, non solo il codice.